Strategic Cost Considerations

Richard P. Rumelt

This technical note covers a number of topics relating to cost that are useful in analyzing business strategies and in making strategic decisions. In particular, it reviews the concepts of economies of scale, economies of scope, and learning-by-doing.

Economies of Scale

It would seem that economies of scale is a simple idea: bigger is better. But the concept is frequently misapplied and misunderstood. Economies of scale exist when larger-scale facilities, optimally used, provide lower average unit costs.

Economies of scale occur with respect to the planned scale or capacity of a production process, not the current rate of production. That is, if a plant operates with annual fixed cost F and variable cost per unit V, then its *average* cost per unit is

$$\bar{C} = V + \frac{F}{Q}. \tag{1}$$

This can be usefully re-written as $\bar{C} = V + F/(Ku)$,where K is capacity and u is the rate of utilization. Let $f = F/K$ be the fixed-cost per unit when operating at full capacity, and note that

$$\bar{C} = V + f/u\,. \tag{2}$$

Define the average cost at full capacity $\ddot{C} = v + f$. This is the quantity which reflects scale economies, if they exist. Combining this with (2) yields

$$\ddot{C} = (1-u)V + u\bar{C} \tag{3}$$

When there are economies of scale, rather than operating leverage, they act to alter $\ddot{C}$, the *average unit cost at full capacity*.

Consider a plant making 50 units per year with fixed (period) costs 1000, variable costs of 10 per unit, and capacity 100 units. In this case, average unit costs are 10 + 1000/50 = 30. Capacity utilization is 50/100 = 50%. The full capacity unit costs $\ddot{C}$ are 10+1000/100 = 20. If production rises to 90, average unit costs fall to 10 + 1000/90 = 21.1. This drop happens simply because the fixed costs are spread over a greater number of units and is sometimes called *operating leverage*.

To discover economies of scale we must change the size or scale of the plant, or the number of plants, not just the rate of production within a plant. Taking this example a step further, suppose that a different facility can be built with a larger capacity of 200. The larger facility has the same variable costs 10 and has fixed expenses 1600. For this larger plant, $\ddot{C}$ = 10 + 1600/200 = 18. This drop from the original plant's $\ddot{C}$ = 20 is evidence of economies of scale. Specifically, this larger facility delivers a 10 percent reduction in average unit cost. In this case the scale economies appear in an increase in fixed expense that was less than proportionate to the increase in capacity.

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Economies of scale arise from a number of phenomena. As the scale of a facility is increased, more specialized equipment can often be designed or purchased, substituting capital for labor on favorable terms. Additionally, the costs of design, construction, and central administration normally do not increase in proportion to scale.

It often requires careful analysis and thought to see if there really are economies of scale in a particular situation. So called "fixed expenses" are often not really fixed, but increase along with the scale of the firm. In many mergers, managers hope for economies of scale from lowering total administrative costs—imagining that these are "fixed expenses" and will not rise in proportion to the size of the firm. However, there is considerable empirical evidence that administrative expenses do not fall with the size of the firm. In fact, the opposite is frequently true. As long as there is a limit to the span of control (the number of people one person can supervise), the number of administrators in a hierarchy must grow faster than the number of bottom level employees!

There may be economies of scale associated with expenses like R&D and advertising. In the case of R&D, there is no particular reason for it to become larger as the size of a single business becomes larger (assuming that the increase in size is not driven by adding more products). Similarly, advertising expenditures may have more to do with the size of the market and competitive moves than the size of the firm.

After some point, scale economies become negligible and diseconomies of scale begin to set in. Existing technology sets an upper limit on scale economies and diseconomies of scale can derive from the increased bargaining power of local suppliers, increased coordination costs among more and more specialized operations, and the tendency of larger organizations to reduce individual motivation.

The Arithmetic of Scale and Cost Dynamics

In manufacturing, the effect of scale on unit cost is usually represented by an exponential function or, equivalently, by a constant elasticity relationship:

$$c = aS^b \tag{4}$$

where S is the scale (or capacity) of the facility, c is the average unit cost of production, b is the scale parameter, and a is a constant—technically the unit cost at a scale of 1. If $b > 0$, then increases in scale increase unit costs—there are diseconomies of scale. By contrast, if $b < 0$, then increases in scale decrease costs and there are economies of scale. [Recall that $S^{-x} = 1/S^x$]. This relationship can also be expressed in terms of ratios:

$$\frac{c_2}{c_1} = \left(\frac{S_2}{S_1}\right)^b \tag{5}$$

This representation says that if scale is changed from S_1 to S_2, then unit costs change from c_1 to c_2. The ratio form eliminates the uninteresting constant a in equation (1).

The scale parameter b is the elasticity of cost with respect to scale. That is, it is the percentage change in cost for a small percentage change in scale. Thus, if $b=-0.3$, then a 1% increase in scale has $c_2/c_1 = 0.997$, a 0.3% drop in unit cost.

It is common practice to refer to the impact of a doubling of scale. We can use equation (2) to determine the impact of doubling scale. Setting, $S_2 = 2S_1$, equation (2) becomes $c_2 = c_1 2^b$. Thus, for example, if $b = -0.3$ then $2^{-0.3} = 0.81$, so a doubling of scale reduces costs to 81% of their former level (a 19% reduction). Conversely, if one is given a scale effect in terms of the doubling effect, it can be converted to b by using the formula

$$b = \frac{\log(c_2 / c_1)}{\log(2)} \tag{6}$$

For example, if doubling scale reduces costs by 10%, then. $b = \log(0.9) / \log(2) = -0.152$. Note that the logarithms can be either natural or common, as long as consistency is maintained.

Exhibit 1—Values of the Scale Coefficient

Value of b	Capital Cost (% of Products)	Operating Cost (% of Products)
	24.4	15.1
-1.0 to -0.51	56.8	56.3
-0.50 to -0.21	13.4	28.1
-0.20 to 0.0	5.4	0.0

A good general guess for scale economies in manufacturing facilities is $b = -0.35$. A study[1] of the U.S. paper industry suggested that doubling uncoated plant paper capacity reduces fixed costs by 28% and labor[2] costs by 32% (the corresponding values of *b* are -0.47 and -0.56 respectively. In a careful (but somewhat dated) study it was found that *b* varied among the products studied as shown in Exhibit 1.

Economies of Scope and Costs of Variety

Economies of scope occur when there are benefits to combining the production of two or more products. For example, in a 1983 study[3] of General Motors, substantial economies of scope were found between large car and small car production—unit costs would have risen by about 25% had these activities been completely separated. By contrast, auto and truck production show no such economies of scope—GM obtained no advantages from producing both automobiles and trucks.

Most scope economies arise from sharing knowledge or from economies of scale in the production of components or services that are utilized by both products. Economies of scope can be often estimated by breaking production down into portions that are common between products and portions that are separable. The economies of scope derive from the extra economies of scale obtained in the combined operation.

The counterpoint to economies of scope are the costs of variety. There is fairly clear evidence that as different activities and products are combined within a facility, costs, especially overhead costs, rise. By analogy to the case of scale economies, the costs of variety can be modeled as

$$C_h = hN^{\nu} \tag{7}$$

[1]Poper, F. *Industry Structure and Competitive Advantage*, Wisconsin, 1984.

[2]Haldi, J. and David W., "Economies of Scale in Industrial Plants, *Journal of Political Economy*, 75, 1967, pp. 373-85.

[3]Friedlaender, A.F., Winston, C. and K. Wang. "Costs, Technology, and Productivity in the U.S. Automobile Industry," *Bell Journal of Economics*, 14, 1983, pp. 1-20.

where c_h is overhead cost, h is a constant, N is the number of different products or services being made, and v is the complexity exponent. Stalk and Hout[4] report complexity slopes for the increase in overhead costs with each doubling in complexity for the products shown in Exhibit 3. (The complexity slope is 2^v.)

In the design of production systems the economies of scale and costs of variety must be considered together. As a simple example, suppose that a single product activity has scale parameter $b=-0.3$ and generates costs of 50/unit at a scale of 1. Thus, variable costs would be described by the scale economies equation $c_V=50S^{-0.3}$. In addition, suppose that as capacity for new product varieties is added each new product requires the same increment in scale and that pure scale effects are achieved for variable costs. But overhead costs per unit, which were one-fifth of variable costs when only one product was produced, rise with variety exponent 0.5. In this case, the complete model is

$$c = 50(NS)^{-0.3} + N^{0.5} \tag{8}$$

where N is the number of products produced. As variety increases, scale effect drive down variable costs but drive up overhead costs. The lowest overall cost is obtained with $N = 3$ products.

Exhibit 2—Variety Effects on Cost

Exhibit 3		
	Complexity	
Product	*Slope*	*Exponent*
Hand Tool Mfg.	154%	0.62
Multinat'l Construction	139%	0.48
Chemical Processing	133%	0.41
Detailed Engineering	121%	0.28
Medical Centers	117%	0.23

Learning by Doing

Average costs tend to fall as an organization gains experience with the manufacture of an object. The effect is called learning-by-doing. It was first studied carefully in 1936 in the airframe industry[5] and has also been carefully documented in shipbuilding,[6] radar equipment,[7] and chemical products.[8] The original "learning curves" applied only to direct labor hours and were of the form

[4]Stalk, G. and Hout, T.M. *Competing Against Time*. New York: The Free Press, 1990, p.48

[5]Wright, T.P. "Factors Affecting the Cost of Airplanes," *Journal of Aeronautical Science*, 3, 1936, pp. 122-28.

[6]Montegomery, Francis J. "Increased Productivity in the Construction of Liberty Vessels," *Monthly Labor Review*, 57, 1943, pp. 861-64.

[7]Preston, L.E. and Keachie, E.C., "Cost Functions and Progress Functions: An Integration," *American Economic Review*, 54, 1964, pp. 100-107.

[8]Lieberman, M.B. "The Learning Curve and Pricing in the Chemical Processing Industries," *Rand Journal of Economics*, 15, 1984, pp. 213-28.

$$L = aQ^{\lambda} \tag{7}$$

where L is the hours of direct labor used in producing the Qth unit, a is the direct labor hours used on the first unit, and 1 is the learning exponent. Note that here Q represents the total cumulative production, not the rate of production. The learning exponent λ is normally negative. During the 1970s the Boston Consulting Group promulgated the idea that learning-by-doing applied not only to direct labor but to all of the non-profit components of value added. This expanded concept was labeled the *experience curve*. The formula for the experience curve is

$$c = aQ^{\lambda} \tag{9}$$

Where c is the average cost net of materials and adjusted for inflation. Again, Q is total cumulative production.

As in the case of scale, equation (8) can also be written in relative terms:

$$\frac{c_2}{c_1} = \left(\frac{Q_2}{Q_1}\right)^{\lambda} \tag{10}$$

and the slope of the experience curve is typically described as 2^{λ}, the fraction of costs remaining after experience (cumulative output) has doubled. This fraction is sometimes called the *progress ratio*. For example, in the semiconductor industry it is common to speak of a 70% progress ratio, implying a learning exponent of

$$\lambda = \log(0.7)/\log(2) = -0.357/0.693 = -0.515.$$

Learning occurs over time and it is thus sometimes useful to look at how growth rates of output and growth rates of cost are interrelated.

- If <u>cumulative</u> output grows at rate g_Q, then the growth rate of cost is λg_Q.
- If output q grows at rate g (per period), then cumulative output grows at rate

$g_Q = \frac{g}{1-(1+g)^{-t}}$, where t is the number of periods since the beginning of production. Thus, if output q grows at a constant rate g, cumulative output has a declining growth rate—the growth rate of cumulate output starts high (much higher than g) but gradually falls closer and closer to g. Thus, in a mature environment (that is, t being large), we would expect 6% annual growth in output together with $\lambda = -0.5$ to yield 3% annual reductions in cost due to learning.

The experience curve is an important concept in strategy because it connects a firm's performance to its history in a way that is difficult for other firms to replicate. Whereas the scale of facilities can be replicated by current investment, its accumulated experience cannot. One study[9] addressed the question of whether learning rates differed among close competitors. The investigators examined cost data on DRAMs and EPROMs supplied to the U.S. Federal Trade Commission by a number Japanese semiconductor makers for the period 1986 to 1991. They found that the learning rates of the firms differed significantly from one another—see Exhibit 3.

[9]Balakrishnan, S. and Udayagiri, N.D. "Firm Differences in Learning Curve Effects." Working Paper, University of Pennsylvania, 1993.

Exhibit 3—Learning Rates		
FIRM	DRAMs	EPROMs
Fujitsu	78%	90%
Hitachi	84%	80%
Toshiba	76%	84%
Mitsubishi	87%	na

An important question is the degree to which learning is transferable from one plant to another and from one product generation to another. In a study of shipbuilding[10] it was found that within the same company, plants starting production seemed to benefit from the experience of existing plants at startup, but that once production got underway there was no further spillover in learning from one plant to another. In a study[11] of a truck manufacturing plant that shifted from one-shift to two-shift operations, it was found that 31% of the learning accumulated during the one-shift era was lost in the move to two-shift operation. In addition, it was found that under two-shift operation, 56% of the learning in each shift benefited the other shift.

A study[12] of learning spillovers in the semiconductor industry, considered as a whole showed differences among products. In DRAMs, there was significant spillover in learning from the previous generation, but not from generations before that. In SRAMs and EPROMs, by contrast, there was significant spillover from all prior generations. This suggests that it would be easier to enter the DRAM business than the EPROM business. Regarding cross-product spillovers, DRAM experience had a significant impact on EPROM and SRAM (slow) productivity, but not on fast SRAMs or EEPROMS. Where the cross-product spillovers were significant, the impact of DRAM experience was roughly equivalent to experience in the product itself.

Where Does Learning-By-Doing Reside?

Learning-by-doing is sometimes thought to be the product of more experienced or practiced workers. A number of lines of research, however, point to the conclusion that learning-by-doing is largely the result of improvements in methods and processes. One particularly cogent study is Mishina's re-examination[13] of the cost data on the B-17 program, which supplied much of the impetus to early theorizing about learning curves. Most of these planes were manufactured in Boeing's main plant (#2) in Seattle, during WWII. Monthly production rose from 5 in September 1941 to 100 by July 1942, peaking at 362 in March 1944. The first 5 aircraft produced consumed 142,837 direct labor hours per aircraft. In August 1944, the average aircraft used 15,316 direct labor hours. Fitting a classical learning curve to the data by regression yields $b = -0.472$, implying in a 28% decline in direct labor hours for each doubling in cumulative production.

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[10]Argote, L.; Beckman, S.L.; and Epple, D. "The Persistance and Transfer of Learning in Industrial Settings," *Management Science*, 36, 1990, pp. 14-54.

[11]Epple, D.; Argote, L.; and Devadas, R. "Organizational Learning Curves: A Method for Investigating Intra-Plant Transfer of Knowledge Acquired Through Learning By Doing," *Organization Science*, 2, 1991, pp. 58-70.

[12]Udayagiri, N.D. and Balakrishnan, S. "Learning Curves and Knowledge Spillovers: The Case of Semiconductor Memories." Working Paper, University of Pennsylvania, 1993.

[13] Kazuhiro Mishina, "Learning by New Experiences: Revisiting the Flying Fortress Learning Curve," in Lamoreau, Raff, and Temin, Learning By Doing, University of Chicago Press, 1999, pp.145-179.

Mishina used extensive historical records to establish that virtually none of this increase could be explained by economies of scale; the basic plant was not altered in scale over this time. Over time, the product became more, rather than less, complex. In addition, the workforce turned over more and more rapidly over time and became *less and less* experienced. By early 1944, much of the workforce had only worked for a year and had never previously held a factory job. Finally, there was no parallel experience at any other facility that might have served as an external source of learning.

Eyewitness accounts to Plant #2 stressed that the increased velocity of production was due to (1) the reduction in re-work, and (2) a reduction in work-in-process inventory. With regard to the first element, re-work was cut though the creation of tools and methods which more completely standardized parts, making them fully interchangeable. With regard to the second element, Plant #2 had moved over time from traditional production to a system without a central stock-room, using small stock-bins at each work-site, and a "pull" system of control. As reported in *Aviation* (July 1943), "Behind all this fluid activity is a perfect timing. If one division falls behind, it is as instantly apparent as an empty space in a line."

Thus, the keys to the increased productivity at Plant #2 were quality methods and a form of JIT ("just in time") production management. Learning resided in the methods and practices used, not in the individual skills of workers. And, interestingly, that learning was both highly specific to the B-17 and fundamentally the same, in abstract terms, as the quality and JIT methods extolled by modern production management gurus.

Decomposing Cost Advantage

It is frequently useful to analyze the sources of cost differences between competitors or potential competitors. Some of the sources of cost difference are easily imitable while others are sustainable despite competition. As an example, suppose that there are two "plants" labeled A and B: **Error! Reference source not found.** summarizes data on the two plants.

Exhibit 4	Plant A	Plant B	Gap
Avg unit cost	62	51	11.000
Raw material cost per unit	12	11	1.000
Avg value-added cost per unit	50	40	10.000
Plant capacity	1000	1350	
Capacity utilization	0.78	0.8	
Fraction variable in Avg cost	0.8	0.83	
Cumulative production experience	15000	20000	
Number of products	16	22	
Cost reduction with double scale	15%	15%	
Cost reduction with double learning	8%	8%	
Cost increase with double variety	8%	8%	

The average costs at the plants are different and the typical assignment is for the analyst to explain the sources of that difference. Doing so means identifying the "drivers" of cost differences—the differences in situation and policy between the firms that create these cost differences. Here, we lack detailed data so the analysis of drivers takes place at a high level of abstraction.

The average unit costs of A are 62 versus 51 at B. Raw material costs are 12 at A and 11 at B, a difference that could come from location, bargaining power, or process technology. Here we simply take it as explaining 1 of the original cost difference of 11, leaving 10 to explain.

In decomposing this gap of 10, we start with the best understood reasons first and only later seek less well-understood drivers to explain the (hopefully) smaller gaps. The clearest reason for a cost difference between these plants is capacity utilization. This effect is often mistakenly called economies of scale, but it is not.

In the case at hand, we are told that variable costs are 80% of total average costs at plant A. Thus, fixed costs are (62)(.2) = 12.4 per unit for A and (.17)(51)= 8.67 for B. As per (2), these numbers are f/u, so that fixed cost per unit at full capacity is (12.4)(.8) = 9.67 for A and (8.67)(.83)=6.94 for B. We don't directly look at this difference right now as it probably reflects scale, learning, and variety effects. Instead, we look at the unit cost penalties of being below capacity levels of operations. At plant A, the penalty is 12.4-9.67=2.73. At B it is 1.734. The difference (0.99) between these penalties is the gain to B from operating leverage. These calculations are summarized below.

Operating Leverage Effect			
Fraction fixed in Avg cost	0.200	0.170	
Avg Fixed Cost per unit	12.400	8.670	
Fixed Cost per Unit at Capacity	9.672	6.936	
Undercapacity penalty	2.728	1.734	0.994

To analyze scale, learning, and variety effects we want to work with full-capacity value-added costs (FCVA). That takes out raw material costs and also takes out the effect of operating leverage. For A, *average variable VA* cost is (.8)(62)-12=37.6 and the comparable figure for B is 31.33. Adding back in *fixed cost at capacity* gives full capacity AV costs of 47.27 and 38.27 for plants A and B. These computations are shown below.

Compute Full Capacity Avg VA Cost		
Avg Variable Cost	49.600	42.330
Avg Variable VA Cost	37.600	31.330
Full Cap VA Cost	47.272	38.266

The three structural sources of cost advantage (or disadvantage) for B over A are scale, learning, and variety. Taking scale, the exponent is log(.85)/log(2)= -0.234. That means that, using (5), if B's scale were only that of A, its costs would increase by (1000/3500)^(-0.234) = 1.073. In other words, if B has A's smaller scale its FCVA would be 7.3% higher or (0.073)(38.27)=$2.79 per unit.

The same logic applied to cumulative learning yields a cost ratio of (20/15)^(-0.1203)=1.035, or a $1.037 per unit increase in B's FCVA. Applying this logic a third time to variety yields a cost ratio of (22/16)^(.111) = 0.965. Thus, if B had A's lower variety, its FCVA would be 3.5% *lower*, or $1.33 per unit.

These computations are summarized in the table below.

	Exponent	Cost ratio for A/B	Advantage to B
Scale	-0.234	1.073	2.790
Learning	-0.120	1.035	1.347
Variety	0.111	0.965	-1.329

Adding up these sources of advantage gives a total estimate of $4.8 per unit compared to the gap of $11. Thus, there are $6.2 in other efficiencies that benefit B. If we had even more data, we could try to decompose that $6.2 into portions due to wage rates, raw material prices, and so on.

Summary of B's Cost Advantage	$/unit	%
Raw Materials	1.000	9.1%
Operational Leverage (capacity util)	0.994	9.0%
Scale	2.790	25.4%
Learning	1.347	12.2%
Variety	(1.329)	-12.1%
Sub Total	4.802	43.7%
Unspecified Efficiencies	6.198	56.3%
Total	11.000	100.0%

This kind of analysis helps shed light on the reasons for cost advantages and dis-advantages. It is not an exact process. Still, the basic shape of the results is what interests the strategy analyst.